CSC 163: Computing, Ethics, and Society

Intellectual Property Disputes: Copyright vs Patent

ABET SO4, performance indicator 4b

Roadmap

1. **The IP landscape** and how copyright differs from patents
2. **Copyright infringement analysis**: ownership, copying, similarity, defenses
3. **Fair use and Google v. Oracle**
4. **Patent infringement analysis**: claims, validity, element-by-element comparison
5. **Worked scenario:** the start-up app
6. **How to answer a 4b-style question**

Learning Objectives (SO4, PI 4b)

After this lecture you can:

- Tell copyright, patent, trade secret and trademark apart
- Apply the **copyright criteria**: valid copyright, copying (access and similarity), substantial similarity of protected expression, defenses
- Apply the **patent criteria**: claims, validity (eligibility, novelty, non-obviousness), every claim element present
- Explain why independent creation defeats copyright but not patent infringement
- Write a clear decision process for a software dispute

The IP Landscape

Four tools, four different kinds of protection

Four Kinds of Intellectual Property

Right	Protects	How obtained	Term (U.S.)	Software example
Copyright	Original expression	Automatic when fixed	Life + 70 years (95 for work made for hire)	Source code, UI graphics, docs
Patent	New, useful, non-obvious inventions	Application and examination	Up to 20 years from filing	A novel compression method
Trade secret	Confidential business information	Keep it secret	While it stays secret	Unreleased code or algorithm
Trademark	Brand identifiers	Use in commerce	Renewable while used	App name and logo

Copyright vs Patent

	Copyright	Patent
Protects	Expression of an idea	An invention as defined in the claims
Obtained by	Creating the work (automatic)	Application, examination, grant by the USPTO
Algorithms and ideas	Not protected	Can be protected if claimed and eligible
Infringement requires	Copying protected expression (access + similarity)	A product that meets **every element** of a claim; copying is not required
Independent creation	A defense	Not a defense
Term	Life + 70 years (individuals)	Up to 20 years from filing

Sources: Copyright Office Circular 61; USPTO patent essentials

Copyright Infringement Analysis

Criteria and steps

Copyright Infringement: Four Steps

1. Valid copyright?

Original work, protectable expression, owned by the claimant

2. Was it copied?

Access to the work plus similarity that copying best explains

3. Substantial similarity?

Compare only the **protected** expression, after filtering out ideas

4. Defenses?

Fair use, license, independent creation, de minimis use

The claimant must win steps 1 to 3; the defendant may still prevail at step 4.

Literal and Non-Literal Copying

- **Literal copying:** identical or near-identical code, files or text
- **Non-literal copying:** the program's **structure, sequence and organization**, screen layout, or user-interface design, even when no line of code is the same
- Non-literal similarity is harder to judge, because much of a program's structure is dictated by function, efficiency or industry conventions
- Courts therefore filter out the unprotectable parts before comparing

The Abstraction-Filtration-Comparison Test

Abstraction

- Break the program into levels, from code up to its purpose

Filtration

- Remove unprotectable material at each level
- Leave the protectable expression

Comparison

- Compare what is left with the accused program
- Is it substantially similar?

Computer Associates v. Altai (2d Cir. 1992): the rewritten program copied no protectable non-literal expression, so no infringement.

What Gets Filtered Out

- **Ideas and processes:** what the program does, not how it is written
- **Efficiency (merger):** when only a few ways of expressing something are practical
- **External factors (scenes a faire):** elements required by hardware, compatibility, industry standards or common programming practice
- **Public domain material:** code that is freely available to everyone

What remains, if anything, is the protectable expression to compare

Fair Use: The Four Factors

Factor	Question	Software illustration
1. Purpose and character	Transformative? Commercial or educational?	Reuse to build a new platform was treated as transformative in Oracle (2021)
2. Nature of the work	Creative or factual and functional?	Functional code sits further from copyright's core
3. Amount used	How much, and how important a part?	About 0.4% of the API in Oracle; the heart of a work can matter
4. Market effect	Does the use substitute for the original?	No loss of actual or potential markets for Java SE in Oracle

No formula: courts weigh the factors together, case by case. Source: U.S. Copyright Office

Google v. Oracle (U.S. Supreme Court, 2021)

What happened

- Google copied about 11,500 lines of declaring code from 37 Java API packages
- It wrote its own code for everything else in Android

What the Court held

- 6 to 2: Google's copying was **fair use**
- The Court assumed, without deciding, that the code was copyrightable

Why it matters

- Fair use is decided by judges, case by case
- Copying only what is needed for a new platform can be fair
- Not a blanket rule for copying code

Patent Infringement Analysis

Claims, validity, and every element

What Can Be Patented

- A utility patent covers a new and useful **process, machine, manufacture, or composition of matter**, or an improvement
- **Not patentable:** laws of nature, physical phenomena, abstract ideas, mere ideas
- Three tests: **utility** (it works), **novelty** (not done before), **non-obviousness** (not an obvious change to prior inventions)
- A patent gives the **right to exclude** others from making, using, selling or importing the invention in the U.S., for up to 20 years from filing

Source: USPTO, patent essentials

Software Patents: The Alice Test

Step 1

Is the claim directed to a patent-ineligible concept, such as an abstract idea?

Step 2

If so, do the claim elements, alone or combined, add an **inventive concept** that makes it eligible?

Alice Corp. v. CLS Bank (2014): implementing an abstract idea (intermediated settlement) on a generic computer was not enough. The Court did not rule out all software patents.

Claims and Infringement

- The **claims** define the invention. Infringement is judged **claim by claim**
- **Literal infringement:** the accused product has **every element** of at least one claim (the all-elements rule)
- **Doctrine of equivalents:** a substitute that does substantially the same function in substantially the same way to get the same result can still infringe
- **No copying needed:** an engineer who invents the same thing independently can still infringe
- **Defenses:** the patent is invalid (prior art, not eligible under Alice, obvious), no infringement, or a license

Patent Infringement: Four Steps

1. Identify the claims

Which patent, which claims, what do the terms mean?

2. Compare elements

Is **every element** of a claim in the product (or an equivalent)?

3. Test validity

Prior art? Eligible under Alice? Obvious?

4. Remedies and options

License, design around, challenge the patent, damages

Related Protections and Ways to Avoid Disputes

- **Trade secret:** taking confidential code or know-how is a separate wrong from copying or patent infringement
- **Clean-room design:** one team writes a specification; a second team, never exposed to the other code, implements it. Supports independent creation
- **Licenses and patent grants:** permissive licenses such as Apache 2.0 include a patent grant from contributors
- **Design around:** change the product so it lacks one claim element

Worked Scenario

The start-up app

The Scenario

A start-up releases a mobile app. Its **user interface**, **source-code structure** and a **key algorithm** closely resemble those of an existing commercial product.

Question: has the start-up violated the other company's **copyright** and/or **patent**?

Plan:

- Treat each element separately: UI, code structure, algorithm
- Use the **copyright criteria** for expression, the **patent criteria** for the invention
- State what evidence you would collect, and what you would conclude if it came back one way or the other

Applying Both Sets of Criteria

Question	Copyright (UI, code, structure)	Patent (key algorithm)
What must the claimant show?	Valid copyright; **copying** (access + similarity); substantial similarity in **protected** expression	A valid patent; **every element** of a claim present in the app
Evidence to collect	Code and UI comparison; access (ex-employees, shared bugs or comments)	Claim chart: each claim element against the app's features
Filtering or validity	Filter ideas, efficiency, standards, public domain	Check prior art, Alice eligibility, obviousness
Defenses	Fair use, license, independent creation	Invalidity, non-infringement, license
Likely turning point	Is the similar part protectable expression?	Is the patent valid and are all elements met?

Element by Element

User interface

- Standard menus and conventions are filtered out
- Distinctive graphics and creative layout may be protected
- Copyright analysis

Code structure

- Run abstraction-filtration-comparison
- Similar structure forced by function is filtered out
- Copyright analysis

Key algorithm

- An algorithm is an idea: **not** covered by copyright
- Only a valid patent, or a trade secret, can protect it
- Patent analysis

How to Answer a 4b-Style Question

1. Name the rights	Say which rights could apply: copyright, patent, maybe trade secret
2. Copyright criteria	Valid copyright, copying, substantial similarity of protected expression, defenses
3. Patent criteria	Claims, validity, every element present, defenses
4. Decide and advise	State the likely outcome and next steps: clean-room rewrite, license, design around

Level 4: correct criteria for both, applied to the scenario, with fair use, clean-room or independent-invention nuances.

Summary

- **Copyright** protects expression and requires **copying**; **patent** protects inventions and does **not** require copying
- Copyright steps: valid copyright, copying, substantial similarity after filtration, defenses
- Patent steps: identify claims, compare every element, test validity, then remedies
- Algorithms and ideas: outside copyright, possibly inside a patent
- Always **separate** the criteria for the two rights in your answer

Exercises and Solutions

Practice for performance indicator 4b

Exercise 1: Which right applies?

Name the main intellectual property right for each item: copyright, patent, trade secret, trademark, or none.

Item	Right
Source code of a mobile app	
A new video compression method described in claims of an issued USPTO patent	
Unreleased recommendation algorithm the company keeps confidential	
The app's name and logo	
Original icons and artwork in the interface	
The bare idea of 'a to-do list app'	

Solution 1: Which right applies?

Copyright: expression. Patent: claimed inventions. Trade secret: confidential know-how. Trademark: brand identifiers. Ideas alone: none.

Item	Right
Source code of a mobile app	**Copyright**
A new video compression method described in claims of an issued USPTO patent	**Patent**
Unreleased recommendation algorithm the company keeps confidential	**Trade secret**
The app's name and logo	**Trademark**
Original icons and artwork in the interface	**Copyright**
The bare idea of 'a to-do list app'	**None (an idea)**

Exercise 2: True or false

Write T or F. If false, correct it with a few words.

Statement	T / F	Correction (if false)
Independent creation is a defense to copyright infringement		
Independent creation is a defense to patent infringement		
Copyright protects an algorithm		
A patent lasts up to 20 years from the filing date		
Fair use is decided by a fixed formula		
To infringe a patent claim, a product must have every element of the claim (literally or by equivalent)		

Solution 2: True or false

1 point for T/F, 1 point for the correction.

Statement	T / F	Correction (if false)
Independent creation is a defense to copyright infringement	T	—
Independent creation is a defense to patent infringement	F	Patent infringement needs no copying; the product only has to meet the claim
Copyright protects an algorithm	F	It protects expression only; an algorithm may be patentable if eligible
A patent lasts up to 20 years from the filing date	T	—
Fair use is decided by a fixed formula	F	Courts weigh four factors case by case
To infringe a patent claim, a product must have every element of the claim (literally or by equivalent)	T	—

Exercise 3: Filtration (Altai)

In the abstraction-filtration-comparison test, unprotectable material is filtered out before comparison. For each element, decide whether it is filtered out or may be protected, and say why.

Element of the program	Filtered or protected?	Reason
The purpose: manage a store's inventory		
A routine written the only practical way to run on the hardware		
Menu layout required by the operating system's standard		
Code copied from a public-domain library		
Original, creative organization of modules with many alternatives		
Original artwork and icons		

Solution 3: Filtration (Altai)

Filtered out: ideas, efficiency (merger), external factors, public domain. Protected expression is what remains.

Element of the program	Filtered or protected?	Reason
The purpose: manage a store's inventory	**Filtered**	**An idea or function**
A routine written the only practical way to run on the hardware	**Filtered**	**Efficiency (merger)**
Menu layout required by the operating system's standard	**Filtered**	**External factors**
Code copied from a public-domain library	**Filtered**	**Public domain**
Original, creative organization of modules with many alternatives	**May be protected**	**Original expression**
Original artwork and icons	**May be protected**	**Original expression**

Exercise 4: Patent claim chart

A patent claim requires: (1) receiving a user's location; (2) retrieving nearby offers; (3) ranking offers by predicted interest; (4) displaying the top offers. Check each product against the claim. Does it literally infringe?

Product	1	2	3	4	Literal infringement?
X: does all four steps	Yes	Yes	Yes	Yes	
Y: lists offers alphabetically, no ranking	Yes	Yes	No	Yes	
Z: all four steps, built independently, team never saw the patent	Yes	Yes	Yes	Yes	

Solution 4: Patent claim chart

Literal infringement needs **all four elements**. Product Y lacks ranking, so there is no literal infringement (the doctrine of equivalents is a separate question). Product Z was built independently, but copying is not needed for patent infringement.

Product	1	2	3	4	Literal infringement?
X: does all four steps	Yes	Yes	Yes	Yes	**Yes: every element present**
Y: lists offers alphabetically, no ranking	Yes	Yes	No	Yes	**No: element 3 missing**
Z: all four steps, built independently, team never saw the patent	Yes	Yes	Yes	Yes	**Yes: independent creation is no defense**

Exercise 5: Practice 4b question

A competitor's photo app has a user interface, code organization and a filter method very similar to yours. Has it violated your **copyright** and/or **patent**? State the criteria you would apply to each right.

Solution 5: Practice 4b question

- **Model answer.**
- Copyright (UI, code organization): I must show a valid copyright, **copying** (access plus similarity), and substantial similarity in **protected** expression after filtering ideas, efficiency, standards and public domain. Defenses: fair use, license, independent creation.
- Patent (filter method): the method is not covered by copyright. Only a valid patent can protect it. I would compare each claim element to the competitor's app; infringement needs **every element**, and independent creation is no defense. The defendant may attack validity (prior art, Alice eligibility, obviousness).
- **Scoring (4 points):** both rights named and separated (1); copyright criteria (1); patent criteria (1); applied to the scenario with evidence and a conclusion or next step such as license or design-around (1).

References

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- Teaching summaries, not legal advice.